

LeetCode 827: Making A Large Island

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Key Insight:

"ab har 0 ko flip kr ke dekhenge , aaju baaju ke saare unique leaders lenge sirf"

"curr = 1; // 0 -> 1 means 1 size increase;"

Disjoint Set Union (DSU) Class Implementation

```
class dsu {
public:

    vector<int> parent;
    vector<int> size;

    dsu(int n){
        parent.resize(n);
        size.resize(n, 1);

        for(int i = 0; i < n; i++){
            parent[i] = i;
        }
    }

    int findpar(int node){
        if(parent[node] == node){
            return node;
        }

        parent[node] = findpar(parent[node]);
        return parent[node];
    }

    void merge(int u, int v){
        int pu = findpar(u);
        int pv = findpar(v);

        if(pu == pv) return;

        if(size[pu] < size[pv]){
            swap(pu, pv);
        }

        size[pu] += size[pv];
        parent[pv] = pu;
    }
};
```



```

class Solution {
public:
    int largestIsland(vector<vector<int>>& grid) {

        int n = grid.size();
        int m = grid[0].size();
        vector<pair<int,int>> dir = {{-1,0}, {0,-1}, {1,0}, {0,1}};
        dsu ds(n * m);

        for(int r = 0; r < n; r++){
            for(int c = 0; c < m; c++){

                if(grid[r][c] == 0) continue;

                int curr = r * m + c;

                for(auto &it : dir){

                    int nr = r + it.first;
                    int nc = c + it.second;

                    if(nr >= 0 && nc >= 0 && nr < n && nc < m){
                        int next = nr * m + nc;

                        if(grid[nr][nc] == 0) continue;
                        ds.merge(curr, next);
                    }
                }
            }
        }

        int ans = 0;
        unordered_set<int> st;
        for(int r = 0; r < n; r++){
            for(int c = 0; c < m; c++){
                if(grid[r][c] == 0) continue;

                int curr = r * m + c;
                int boss = ds.findpar(curr);
                st.insert(boss);
            }
        }

        for(auto &it : st){
            ans = max(ans, ds.size[it]);
        }

        // ab har 0 ko flip kr ke dekhenge , aaju baaju ke saare unique leaders lenge sirf
        for(int r = 0; r < n; r++){

```

```

    for(int c = 0; c < m; c++){

        if(grid[r][c] == 1) continue;

        unordered_set<int> st;
        for(auto &it : dir){

            int nr = r + it.first;
            int nc = c + it.second;

            if(nr >= 0 && nc >= 0 && nr < n && nc < m){

                int next = nr * m + nc;
                if(grid[nr][nc] == 1){
                    int boss = ds.findpar(next);
                    st.insert(boss);
                }
            }
        }

        int curr = 1; // 0 -> 1 means 1 size increase;

        for(auto &it : st){
            int toadd = ds.size[it];
            curr += toadd;
        }

        ans = max(ans, curr);
    }

    return ans;
}

};

```